

TASK – 2

SIEM – ATTACK SIMULATION AND DETECTION

1. Introduction

This project focuses on creating logs, correlation rules, and testing them with Elastic Stack SIEM lab using Elasticsearch, Kibana, LogStash and Filebeat for log creation, collection, detection and visualization of attacks like credential stuffing, DNS tunnelling and PowerShell exploitation through centralized log monitoring and real-time analysis.

2. Objectives

- To set up and configure the ELK Stack in a virtualized environment.
- To create custom logs, correlation rules and test them on an SIEM platform.
- To collect and analyse security logs from Linux systems.
- To detect attacks like credential stuffing, DNS tunnelling and PowerShell exploitation.
- To generate alerts and visualize suspicious behaviour through Kibana dashboards.
- To get hands-on experience on real-time threat analysis and understand how SOC Analysts monitor, detect and investigate them.

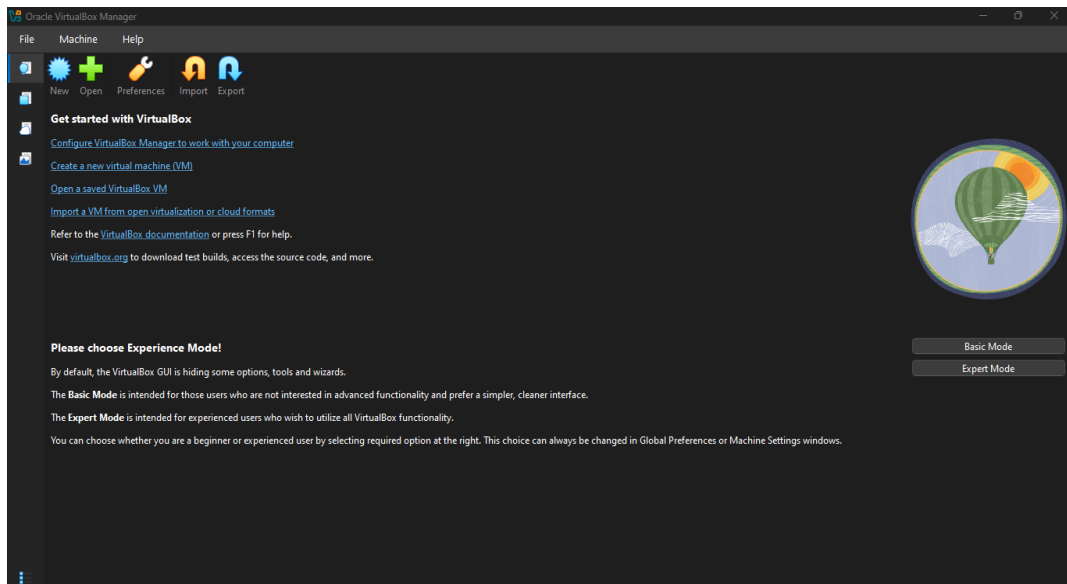
3. Tools Used

- ELK Stack
 - E – Elasticsearch (log storage and indexing).
 - L – LogStash (fake log creation, generation and ingestion).
 - K – Kibana (visualization, log search and alert monitoring).
- Ubuntu ISO – hosts the ELK Stack environment.
- Oracle VM Virtual Box – creates and runs virtual machines.
- FileBeat – sends custom logs to Elasticsearch for analysis.
- Ubuntu Terminal – updates ubuntu packages, installs java, Elasticsearch, Kibana.
- Kali Linux - attack simulation and testing.

4. Lab Setup

1) Install Virtual Box

Oracle VM Virtual Box creates and runs virtual machines.

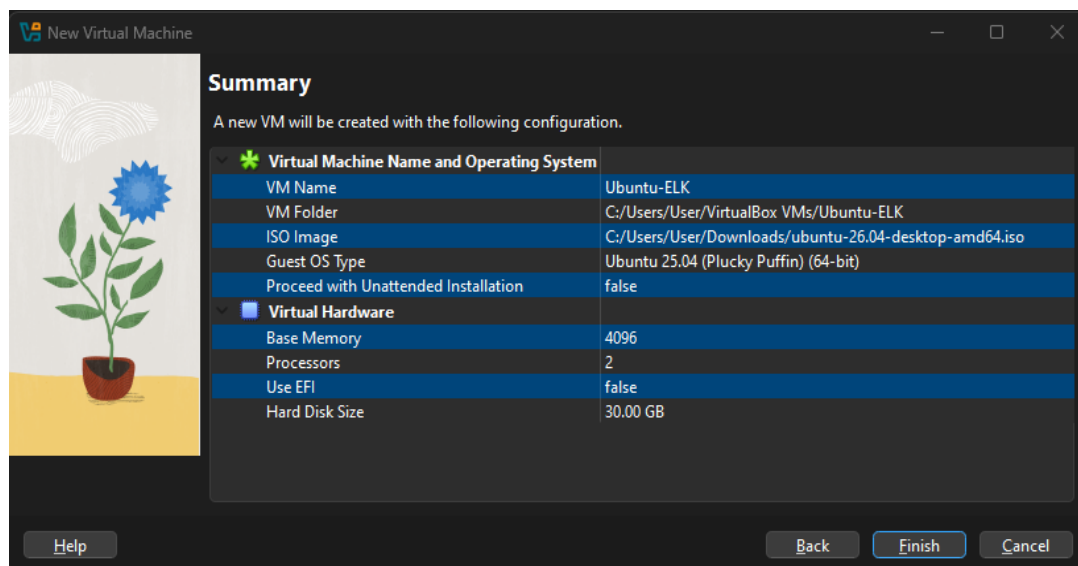


2) Install Ubuntu ISO

It installs Ubuntu in the created virtual computer from the previous step.

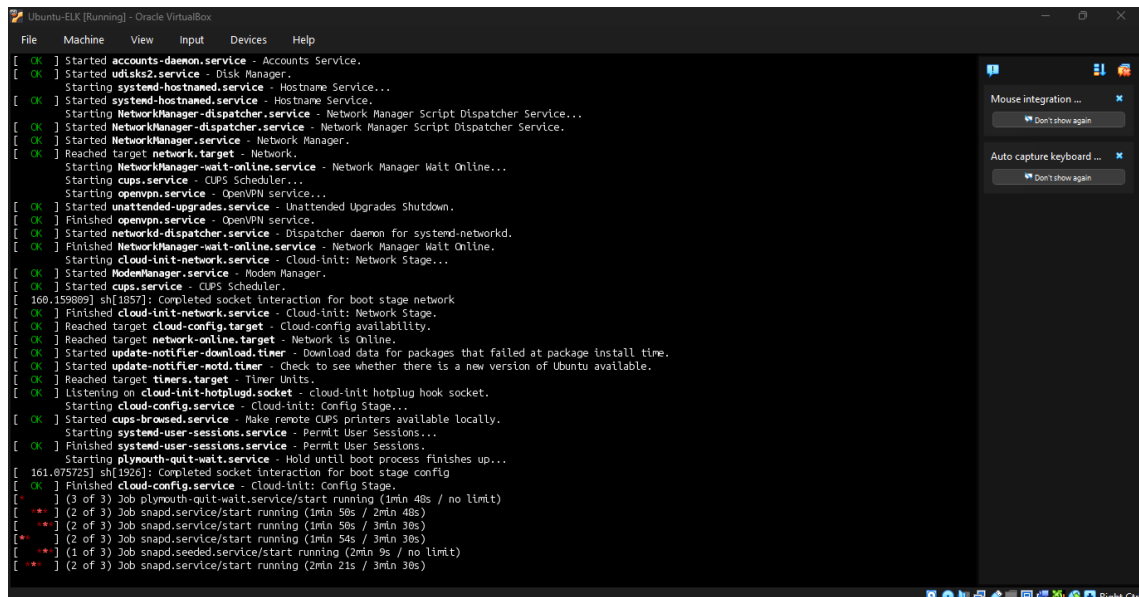
3) Create VM

Create a virtual machine in the virtual box.



4) Start Ubuntu Installation

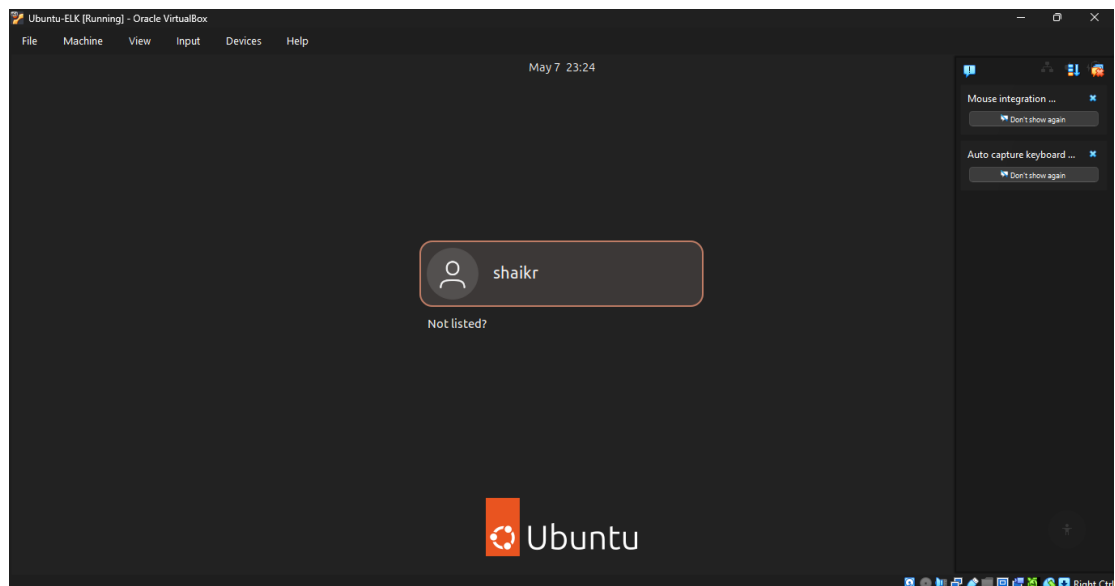
Select Ubuntu ELK and Start.



The screenshot shows a terminal window titled "Ubuntu-ELK [Running] - Oracle VirtualBox". The terminal displays a series of system logs indicating the successful startup of various services. The logs include messages for starting services like accounts-daemon.service, udisk2.service, systemd-hostnamed.service, NetworkManager-dispatcher.service, NetworkManager.service, NetworkManager-wait-online.service, cups.service, openvpn.service, unattended-upgrades.service, openvpn.service, networkd-dispatcher.service, NetworkManager-wait-online.service, cloud-init-network.service, ModemManager.service, cups.service, and plymouth-quit-wait.service. The logs also show the completion of the boot stage network, cloud-init-network, and cloud-config.target, as well as the start of the update-notifier-notd.timer and update-notifier-notd.timer. The terminal output is as follows:

```
[ OK ] Started accounts-daemon.service - Accounts Service.
[ OK ] Started udisk2.service - Disk Manager.
[ OK ] Starting systemd-hostnamed.service - Hostname Service...
[ OK ] Started systemd-hostnamed.service - Hostname Service.
[ OK ] Starting NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service...
[ OK ] Started NetworkManager-dispatcher.service - Network Manager Script Dispatcher Service.
[ OK ] Started NetworkManager.service - Network Manager.
[ OK ] Reached target network.target - Network.
[ OK ] Starting NetworkManager-wait-online.service - Network Manager Wait Online...
[ OK ] Starting cups.service - CUPS Scheduler...
[ OK ] Starting openvpn.service - OpenVPN service...
[ OK ] Started unattended-upgrades.service - Unattended Upgrades Shutdown.
[ OK ] Finished openvpn.service - OpenVPN service.
[ OK ] Started networkd-dispatcher.service - Dispatcher daemon for systemd-networkd.
[ OK ] Finished NetworkManager-wait-online.service - Network Manager Wait Online.
[ OK ] Starting cloud-init-network.service - Cloud-Init: Network Stage...
[ OK ] Started ModemManager.service - Modem Manager.
[ OK ] Started cups.service - CUPS Scheduler.
[ OK ] 160.159809] sh[1857]: Completed socket interaction for boot stage network.
[ OK ] Finished cloud-init-network.service - Cloud-Init: Network Stage.
[ OK ] Reached target cloud-config.target - Cloud-config availability.
[ OK ] Reached target network-online.target - Network is Online.
[ OK ] Started update-notifier-notd.timer - Download data for packages that failed at package install time.
[ OK ] Started update-notifier-notd.timer - Check to see whether there is a new version of Ubuntu available.
[ OK ] Reached target timers.target - Timer Units.
[ OK ] Listening on cloud-init-hotplugd.socket - cloud-init hotplug hook socket.
[ OK ] Starting cloud-config.service - Cloud-Init: Config Stage...
[ OK ] Started cups-browsed.service - Make remote CUPS printers available locally.
[ OK ] Starting systemd-user-sessions.service - Permit User Sessions...
[ OK ] Finished systemd-user-sessions.service - Permit User Sessions.
[ OK ] Starting plymouth-quit-wait.service - Hold until boot process finishes up...
[ OK ] 161.075725] sh[1925]: Completed socket interaction for boot stage config.
[ OK ] Finished cloud-config.service - Cloud-Init: Config Stage.
[ OK ] (3 of 3) Job plymouth-quit-wait.service/start running (1min 48s / no limit)
[ OK ] (2 of 3) Job snapd.service/start running (1min 48s / 2min 48s)
[ OK ] (2 of 3) Job snapd.service/start running (1min 50s / 3min 30s)
[ OK ] (2 of 3) Job snapd.service/start running (1min 54s / 3min 30s)
[ OK ] (1 of 3) Job snapd.seeded.service/start running (2min 9s / no limit)
[ OK ] (2 of 3) Job snapd.service/start running (2min 21s / 3min 30s)
```

Create account and install Ubuntu.



Now, the Ubuntu VM setup is fully complete.

5. Install Tool

1) Install ELK Stack

Update Ubuntu packages.

```

shaikr@shaikr-VirtualBox:/etc/apt/sources.list.d$ sudo apt upgrade -y
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-modules-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-tools-7.0.0-14
  linux-image-unsigned-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-main-modules-zfs-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

Get another security update through Ubuntu Pro with 'esm-apps' enabled:
  libopenexr-3-1-30
Learn more about Ubuntu Pro at https://ubuntu.com/pro
Upgrading:
  distro-info-data  python3-pil

```

2) Install Java

For Elasticsearch to work in the VM.

```

shaikr@shaikr-VirtualBox:/etc/apt/sources.list.d$ java -version
openjdk version "11.0.30" 2026-01-20
OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu1)
OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu1, mixed mode, sharing)
shaikr@shaikr-VirtualBox:/etc/apt/sources.list.d$

```

3) Install Elasticsearch

The main tool for performing the project on an SIEM platform.

```

shaikr@shaikr-VirtualBox: ~
(Reading database... 196711 files and directories currently installed.)
Preparing to unpack .../curl_8.18.0-1ubuntu2.1_amd64.deb...
Unpacking curl (8.18.0-1ubuntu2.1)...
Setting up curl (8.18.0-1ubuntu2.1)...
Processing triggers for man-db (2.13.1-1build1)...
shaikr@shaikr-VirtualBox:~$ curl http://localhost:9200
{
  "name" : "shaikr-VirtualBox",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "MDIRVNCgTUCmNs1COFRkoA",
  "version" : {
    "number" : "7.17.28",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "139cb5a961d8de68b8e02c45cc47f5289a3623af",
    "build_date" : "2025-02-20T09:05:31.349013687Z",
    "build_snapshot" : false,
    "lucene_version" : "8.11.3",
    "minimum_wire_compatibility_version" : "6.8.0",
    "minimum_index_compatibility_version" : "6.0.0-beta1"
  },
  "tagline" : "You Know, for Search"
}
shaikr@shaikr-VirtualBox:~$

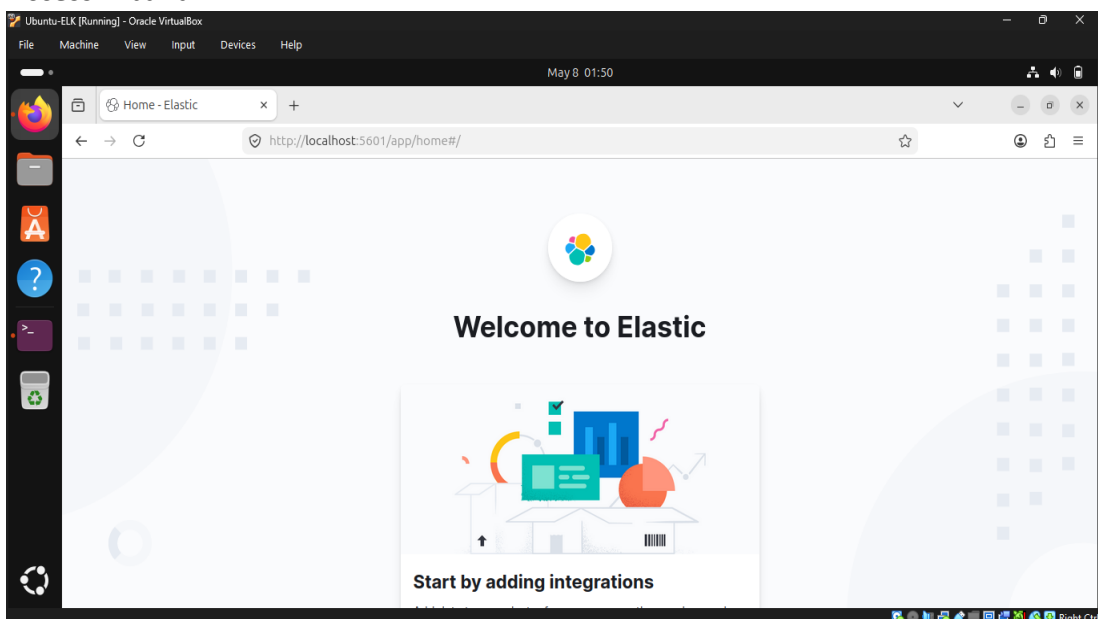
```

4) Install Kibana

Used for visualization and creating the dashboards.

```
shaikr@shaikr-VirtualBox: ~  
Created Kibana keystore in /etc/kibana/kibana.keystore  
shaikr@shaikr-VirtualBox:~$ sudo systemctl enable kibana  
Synchronizing state of kibana.service with SysV service script with /usr/lib/sys  
temd/systemd-sysv-install.  
Executing: /usr/lib/systemd/systemd-sysv-install enable kibana  
Created symlink '/etc/systemd/system/multi-user.target.wants/kibana.service' ->  
'/etc/systemd/system/kibana.service'.  
shaikr@shaikr-VirtualBox:~$ sudo systemctl start kibana  
shaikr@shaikr-VirtualBox:~$ sudo systemctl status kibana  
● kibana.service - Kibana  
   Loaded: loaded (/etc/systemd/system/kibana.service; enabled; preset: enabl  
   Active: active (running) since Fri 2026-05-08 01:46:00 IST; 8s ago  
 Invocation: db8634b5cb5a4c4e9d5aeae9288566d2  
    Docs: https://www.elastic.co  
   Main PID: 7662 (node)  
     Tasks: 7 (limit: 4024)  
    Memory: 94.3M (peak: 94.3M)  
       CPU: 7.060s  
    CGroup: /system.slice/kibana.service  
            └─7662 /usr/share/kibana/bin/./node/bin/node /usr/share/kibana/bi>  
  
May 08 01:46:00 shaikr-VirtualBox systemd[1]: Started kibana.service - Kibana.  
May 08 01:46:00 shaikr-VirtualBox kibana[7662]: Kibana is currently running with  
shaikr@shaikr-VirtualBox:~$
```

5) Access Kibana



6) Install LogStash

```

shaikr@shaikr-VirtualBox:~$ sudo systemctl status logstash
● logstash.service - logstash
   Loaded: loaded (/etc/systemd/system/logstash.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-05-14 14:22:51 IST; 39s ago
     Invocation: 0fe1d5d7ac9d4691b9e9d46aef7e8d87
       Main PID: 4286 (java)
         Tasks: 15 (limit: 1859)
        Memory: 440.2M (peak: 440.2M, swap: 3.2M, swap peak: 3.2M)
           CPU: 33.377s
        CGroup: /system.slice/logstash.service
                └─4286 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -XX:+UseConcMarkSweepGC -

```

7) Install and configure Filebeat.

```

shaikr@shaikr-VirtualBox: ~ -- sudo nano /etc/filebeat/filebeat.yml
GNU nano 8.7.1 /etc/filebeat/filebeat.yml
##### Filebeat Configuration Example #####

# This file is an example configuration file highlighting only the most common
# options. The filebeat.reference.yml file from the same directory contains all the
# supported options with more comments. You can use it as a reference.
#
# You can find the full configuration reference here:
# https://www.elastic.co/guide/en/beats/filebeat/index.html
#
# For more available modules and options, please see the filebeat.reference.yml sample
# configuration file.

##### Filebeat inputs #####

filebeat.inputs:
- type: log
  id: custom-attack-logs
  enabled: true
  paths:
    - /home/shaikr/siem-project/powershell.log
    - /home/shaikr/siem-project/dns.log
    - /home/shaikr/siem-project/credential.log

```

```

shaikr@shaikr-VirtualBox:~$ sudo filebeat test output
elasticsearch: http://localhost:9200...
parse url... OK
connection...
  parse host... OK
  dns lookup... OK
  addresses: 127.0.0.1
  dial up... OK
TLS... WARN secure connection disabled
talk to server... OK
version: 7.17.28
shaikr@shaikr-VirtualBox:~$

```

6. Attack Simulation

I. Attack Overview

ATTACK 1 – Credential Stuffing

This attack is where attackers use stolen usernames and passwords from previous data breaches to try logging into others' accounts, which if still exists or is unchanged, allows them to gain unauthorized access.

ATTACK 2 – DNS Tunnelling

- This is an attacker technique where they misuse DNS protocol for secret data transfer with the compromised systems. And since it's a commonly used protocol, attackers may hide malicious communication in the DNS queries.

- Usually used for C2 communication or data exfiltration.

ATTACK 3 — PowerShell Exploitation

This is a cyberattack where attackers misuse Windows PowerShell to download malicious payloads, execute malicious commands or maintain persistence.

Since PowerShell is legitimate, often used as a means for malicious activities.

II. Create project folder.

```
shaikr@shaikr-VirtualBox:~$ mkdir ~/siem-project
shaikr@shaikr-VirtualBox:~$ cd ~/siem-project
shaikr@shaikr-VirtualBox:~/siem-project$
```

III. Log Creation

Create fake attack logs.

ATTACK 1 – Credential Stuffing

```
$ sudo /usr/share/logstash/bin/logstash -f /etc/logstash/conf.d/generator.conf
```

Logstash Config

```
shaikr@shaikr-VirtualBox: ~ — sudo nano /etc/logstash/conf.d/generator.conf
GNU nano 8.7.1 /etc/logstash/conf.d/generator.conf
input{
generator{
lines => [
'{"source.ip":"192.168.1.100","http.response.status_code":401,"message":"Failed login","user":"admin"}'
]
count => 0
}
}
filter{
json{
source => "message"
}
mutate{
remove_field => ["message","host","@version"]
}
}
output{
elasticsearch{
hosts => ["http://localhost:9200"]
index => "failed-logins-%{+YYYY.MM.dd}"
}
stdout{
codec => rubydebug
}
}
```

ATTACK 2 – DNS Tunnelling

```
shaikr@shaikr-VirtualBox:~$ sudo /usr/share/logstash/bin/logstash -f /etc/logstash/conf.d/dns-generator.conf
```

Logstash Config

```
shaikr@shaikr-VirtualBox: ~ -- sudo nano /etc/logstash/conf.d/dns-generator.conf

GNU nano 8.7.1 /etc/logstash/conf.d/dns-generator.conf *
input{
  generator{
    lines => [
      '{"source.ip":192.168.1.120,"dns.question.name":"verylongdomain12345678.malicious-domain.com","message":"Suspicious DNS query"}'
    ]
    count => 0
  }
}
filter{
  json{
    source => "message"
  }
}
mutate{
  remove_field => ["message","host","@version"]
}
}
output{
  elasticsearch{
    hosts => ["http://localhost:9200"]
    index => "dns-tunnel-%{+YYYY.MM.dd}"
  }
}
stdout{
  codec => rubydebug
}
}
```

ATTACK 3 — PowerShell Exploitation

```
shaikr@shaikr-VirtualBox:~$ sudo nano /etc/logstash/conf.d/powershell-generator.conf
```

Logstash Config

```
shaikr@shaikr-VirtualBox: ~ -- sudo nano /etc/logstash/conf.d/powershell-generator.conf

GNU nano 8.7.1 /etc/logstash/conf.d/powershell-generator.conf *
input{
  generator{
    lines => [
      '{"host.name":"windows-victim","process.name":"powershell.exe","powershell.command":"powershell -enc test","message":"PowerShell EncodedCommand detected"}'
    ]
    count => 0
  }
}
filter{
  json{
    source => "message"
  }
}
mutate{
  remove_field => ["message","host","@version"]
}
}
output{
  elasticsearch{
    hosts => ["http://localhost:9200"]
    index => "powershell-logs-%{+YYYY.MM.dd}"
  }
}
stdout{
  codec => rubydebug
}
}
```

IV. Log Collection

Collection of the created fake logs to forward to elasticsearch and kibana for further attack process.

ATTACK 1 – Credential Stuffing

```
shaikr@shaikr-VirtualBox:~$ curl http://localhost:9200/_cat/indices?v | grep failed-logins
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload  Total   Spent    Left    Speed
100 1300 100 1300  0    0 11067      0 --:--:-- --:--:-- --:--:-- 0
yellow open failed-logins-2026.05.13      bHoXmrw0TwubIxt0D5jG5g  1  1    44876      0    1.4mb    1.4mb
shaikr@shaikr-VirtualBox:~$
```



```

        "@timestamp" => 2026-05-14T09:32:46.983Z,
        "source.ip" => "192.168.1.100",
        "sequence" => 13633
    }
    {
        "http.response.status_code" => 401,
        "user" => "admin",
        "@timestamp" => 2026-05-14T09:32:46.983Z,
        "source.ip" => "192.168.1.100",
        "sequence" => 13634
    }
    {
        "http.response.status_code" => 401,
        "user" => "admin",
        "@timestamp" => 2026-05-14T09:32:46.986Z,
        "source.ip" => "192.168.1.100",
        "sequence" => 13635
    }
    {
        "http.response.status_code" => 401,
        "user" => "admin",
        "@timestamp" => 2026-05-14T09:32:46.986Z,
        "source.ip" => "192.168.1.100",
        "sequence" => 13636
    }
    {
        "http.response.status_code" => 401,
        "user" => "admin",
        "@timestamp" => 2026-05-14T09:32:46.987Z,
        "source.ip" => "192.168.1.100"
    }

```

ATTACK 2 – DNS Tunnelling

```

{
    "@timestamp" => 2026-05-14T09:45:21.123Z,
    "event.category" => "network",
    "event.type" => "info",
    "message" => "Suspicious DNS query detected: a1b2c3d4ef.example.com",
    "source.ip" => "192.168.1.100",
    "query.name" => "a1b2c3d4ef.example.com",
    "dns.qtype" => "A",
    "sequence" => 22345
}
{
    "@timestamp" => 2026-05-14T09:45:24.456Z,
    "event.category" => "network",
    "event.type" => "info",
    "message" => "Suspicious DNS query detected: x9y8z7w6vu.example.com",
    "source.ip" => "192.168.1.100",
    "query.name" => "x9y8z7w6vu.example.com",
    "dns.qtype" => "A",
    "sequence" => 22346
}
{
    "@timestamp" => 2026-05-14T09:45:27.789Z,
    "event.category" => "network",
    "event.type" => "info",
    "message" => "Suspicious DNS query detected: k7j6h5g4fd.example.com",
    "source.ip" => "192.168.1.100",
    "query.name" => "k7j6h5g4fd.example.com",
    "dns.qtype" => "A",
    "sequence" => 22347
}

```

ATTACK 3 — POWERSHELL EXPLOITATION

```
{
  "@timestamp" => 2026-05-14T10:05:11.321Z",
  "event.category" => "process",
  "event.type" => "start",
  "message" => "PowerShell process started:whoami",
  "process.name" => "powershell.exe",
  "process.command_line" => "powershell -Command whoami",
  "source.ip" => "192.168.1.100",
  "user.name" => "admin",
  "sequence" => 33451
}
{
  "@timestamp" => 2026-05-14T10:05:18.654Z",
  "event.category" => "process",
  "event.type" => "start",
  "message" => "PowerShell process started:EncodedCommand detected",
  "process.name" => "powershell.exe",
  "process.command_line" => "powershell -EncodedCommand SQBFAFgA",
  "source.ip" => "192.168.1.100",
  "user.name" => "admin",
  "sequence" => 33452
}
{
  "@timestamp" => 2026-05-14T10:05:24.987Z",
  "event.category" => "process",
  "event.type" => "start",
  "message" => "PowerShell process started:Invoke-Expression",
  "process.name" => "powershell.exe",
  "process.command_line" => "powershell -Command Invoke-Expression (New-Object Ne
  "source.ip" => "192.168.1.100",
  "user.name" => "admin",
  "sequence" => 33453
}
```

V. Log Ingestion

Logstash send logs to elastic search indexes.

```
INFO [2026-05-14 15:01:31.799] [[main]-pipeline-manager] elasticsearch - Elasticsearch pool URLs updated (:changes=>{:removed=>[]}, :added=>[http://localhost:9200/])
WARN [2026-05-14 15:01:35.330] [[main]-pipeline-manager] elasticsearch - Restored connection to ES instance {:url=>"http://localhost:9200/"}
INFO [2026-05-14 15:01:35.560] [[main]-pipeline-manager] elasticsearch - Elasticsearch version determined (7.17.20) {:es_version=>7}
WARN [2026-05-14 15:01:35.606] [[main]-pipeline-manager] elasticsearch - Detected a 6.x and above cluster: the 'type' event field won't be used to determine the document type {:es_version=>7}
```

Kibana → Stack Management → Index Patterns → Create

ATTACK 1 – Credential Stuffing

Create index pattern

Name

failed-logins-*

Use an asterisk (*) to match multiple characters. Spaces and the characters , / , ? , " , < , > , | are not allowed.

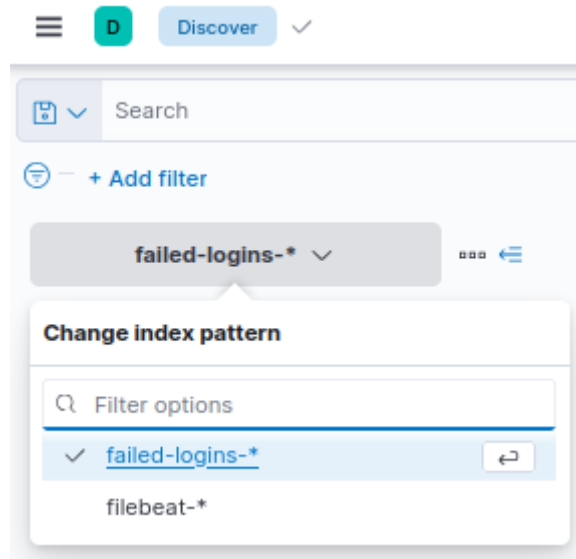
Timestamp field

@timestamp |

Select a timestamp field for use with the global time filter.

[Show advanced settings](#)

Created an index pattern, **failed-logins-***.
Look for the new index created in Discover section.



ATTACK 2 – DNS Tunnelling

Index patterns

Create and manage the index patterns that help you retrieve your data from Elasticsearch.



Created an index pattern, **dns-tunnel**.

ATTACK 3 — POWERSHELL EXPLOITATION

powershell-*



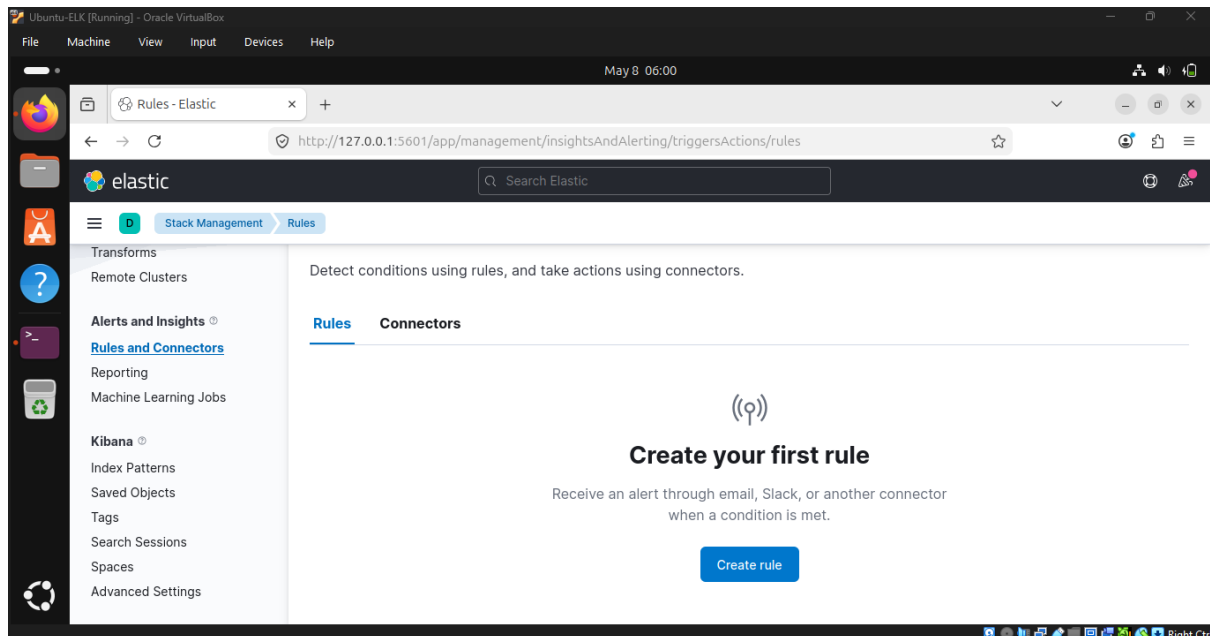
View and edit fields in **powershell-***. Field attributes, such as type and searchability, are based on [field mappings](#) in Elasticsearch.

Fields (13) Scripted fields (0) Field filters (0)					
Search All field types Add field					
Name ↑	Type	Format	Searchable	Aggregatable	Excluded
_id	_id		•	•	
_index	_index		•	•	
...					

Created an index pattern, **powershell-***.

8. Create Correlation Rules

In Kibana, go to **Stack Management -> Rules and Connectors -> Create Rule**



Rule 1- Credential Stuffing Detection

This can indicate brute-force attempts where attackers repeatedly use different combinations of credentials to gain unauthorized access.

1) Set rule threshold and values.

SIZE 100

Define the Elasticsearch query

Elasticsearch query

```
1 {
2   "query": {
3     "match": {
4       "message": "Failed login"
5     }
6   }
7 }
```

[Elasticsearch Query DSL documentation](#)

[Test query](#)

When number of matches

IS ABOVE 4

FOR THE LAST 30 minutes

Looks for queries matching the elastic query message set from **failed-logins-***, after checking every 1 minute, detects and notifies when there are 4 attempts as the threshold is set.

2) Create Connector

✕

🔧

Edit connector

Configuration

Test

Connector name

credential

Connector settings

Write to index

Index

cred-alerts

✕

Use * to broaden your query.

[Configuring index connector.](#)

☒

Refresh index

?

Cancel

Save

Save & close

3) Add Action for triggering alerts.

Rules and Connectors

Detect conditions using rules, and take actions using connectors.

Rules

Connectors

Create rule

Search

Showing: 3 of 3 rules.

Active: 0

Error: 0

Ok: 3

Pending: 0

Unknown: 0

☐	Enabled	Name ↑	Last run	Interval
☐	☒	Credential stuffing detection Elasticsearch query	May 14, 2026 14:35:42pm a few seconds ago	1 min
☐	☒	DNS Tunneling detection Elasticsearch query	May 14, 2026 14:35:42pm a few seconds ago	1 min
☐	☒	PowerShell Exploitation detection Elasticsearch query	May 14, 2026 14:35:42pm a few seconds ago	1 min

Rows per page: 10

✕

Edit rule

Actions

credential

✕

Run when

Query matched

▼

Index connector

[Add connector](#)

credential

▼

Document to index

🔗

```

1 {
2   "alert_type": "credential stuffing attack",
3   "status": "Triggered",
4   "message": "Failed login",
5   "timestamp": "{{date}}",
6   "rule_name": "{{rule.name}}",
7   "severity": "high"
8 }

```

Cancel

Save

Rule 2- DNS Tunneling detection

Unusual sub-domains may indicate encoded data being shared through DNS queries.

1) Set rule threshold and values.

SIZE 100

Define the Elasticsearch query

Elasticsearch query

```

1 {
2   "query": {
3     "match": {
4       "message": "Suspicious DNS"
5     }
6   }
7 }

```

[Elasticsearch Query DSL documentation](#)

[Test query](#)

When number of matches

IS ABOVE 2

FOR THE LAST 30 minutes

Looks for queries matching the elastic query message set from the dns-tunnel, after checking every 1 minute, detects and notifies when there are 2 attempts as the threshold is set.

2) Create Connector



Index connector
×

Index data into Elasticsearch.

Connector settings

Write to index

Index

×

Use * to broaden your query.

[Configuring index connector.](#)

☒ Refresh index

[Back](#)
[Save & test](#)
[Save](#)

3) Add Action for triggering alerts.

Edit rule



Actions

▼

 dnstunel



Run when

Query matched

▼

Index connector

Add connector

dnstunel

▼

Document to index



1

Cancel

✓ Save

Rule 3 – PowerShell Exploitation detection

Detects suspicious PowerShell commands executed commonly used for malware delivery and execution, like EncodedCommand, whoami.

1) Set rule threshold and values.

SIZE 100

Define the Elasticsearch query

Elasticsearch query

```
1 {
2   "query": {
3     "match": {
4       "message": "whoami"
5     }
6   }
7 }
8
```

[Elasticsearch Query DSL documentation](#)

[Test query](#)

When number of matches

IS ABOVE 1

FOR THE LAST 60 minutes

Looks for queries matching the elastic query message set from the powershell-*, after checking every 1 minute, detects and notifies when there is 1 attempt as the threshold is set.

2) Create Connector


Index connector
×

Connector name

powershell

Connector settings

Write to index

Index

powershell-alert

Use * to broaden your query.

[Cancel](#)
[✓ Save](#)

3) Add Action for triggering alerts.

Rules and Connectors

Detect conditions using rules, and take actions us

Rules **Connectors**

Search...

- ☐ Name Ty
- ☐ credential In
- ☐ dnstunel In
- ☐ dnstunnel In


Index connector
×

Index data into Elasticsearch.

Connector settings

Write to index

Index

powershell-logs

Use * to broaden your query.

[Configuring index connector.](#)

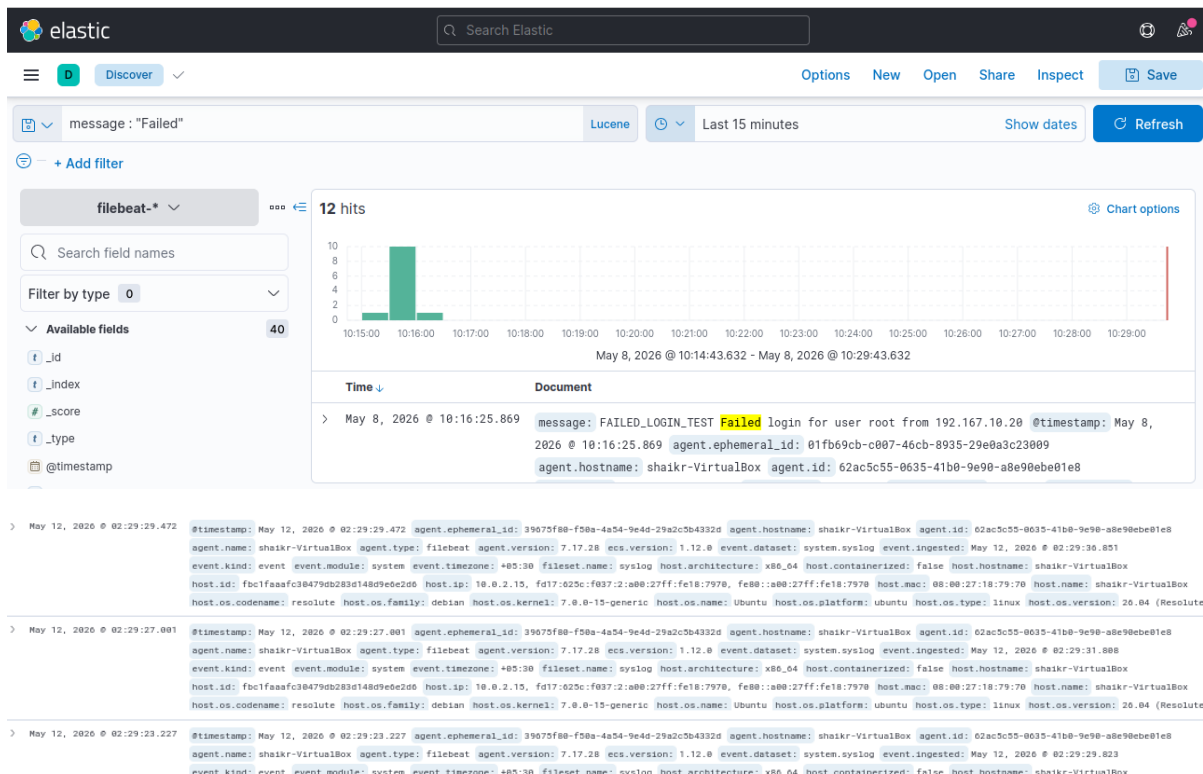
☒ Refresh index

[Back](#)
[Save & test](#)
[Save](#)

9. Detection Analysis

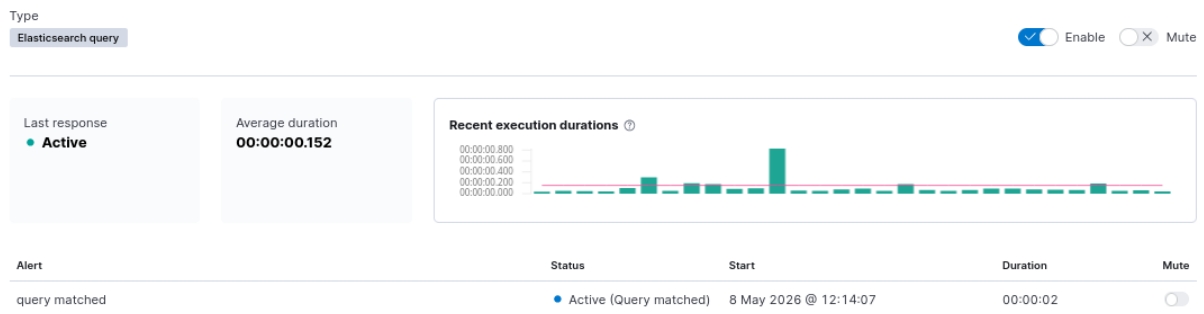
The simulated attack logs were successfully ingested to Kibana. Each rule matched its respective detection keywords, thresholds and successfully generated alerts. The active states of alerts show successful rule execution confirming working of the logic.

Rule Name	Query Logic	Threshold	Purpose
Credential Stuffing	message: "Failed login"	4 events	Multiple failed login attempts
DNS Tunnelling	message: "Suspicious DNS query"	2 events	Suspicious DNS queries detected
PowerShell Exploitation	message: "PowerShell"	1 event	Suspicious PowerShell command usage detected



ATTACK 1 – Credential Stuffing

Credential stuffing detection



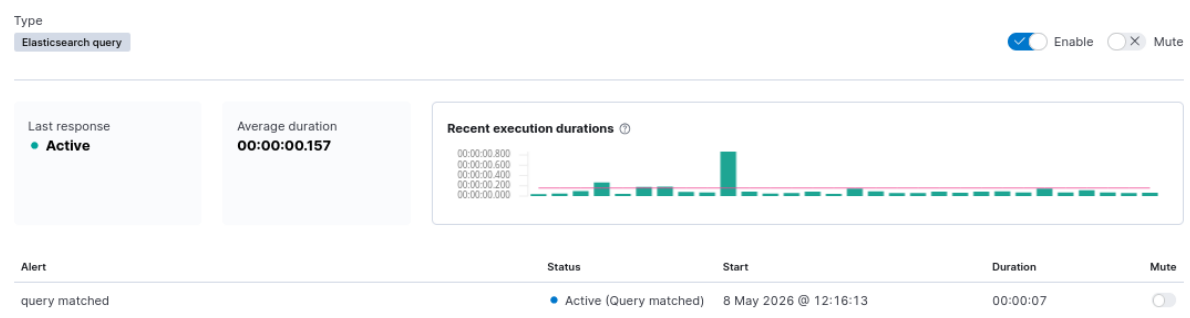
ATTACK 2 – DNS Tunnelling

DNS Tunnelling detection



ATTACK 3 — PowerShell Exploitation

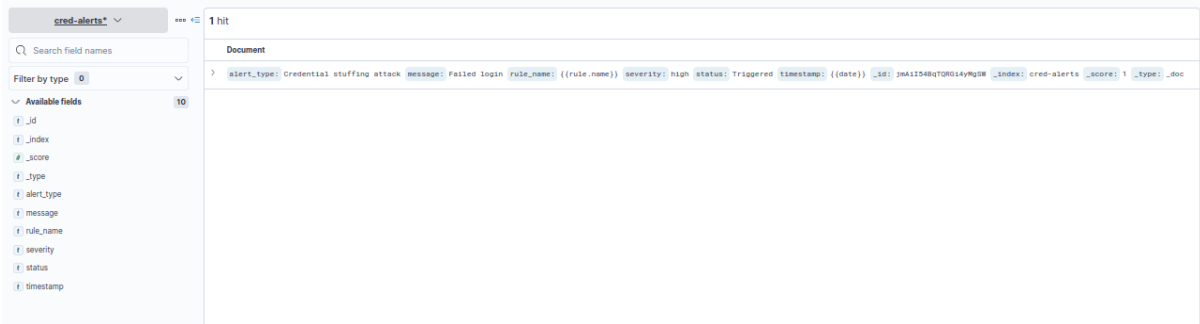
PowerShell Exploitation detection



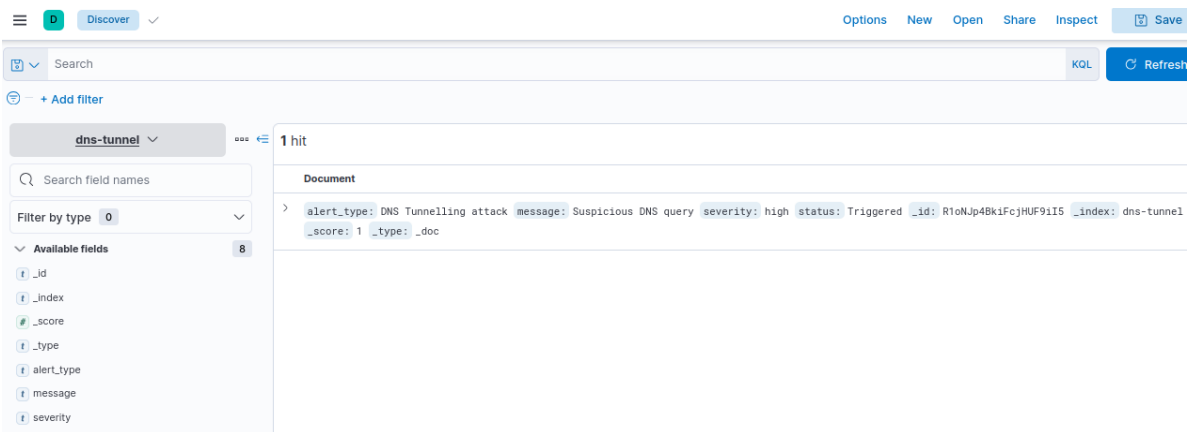
10. Alerts Generated

Look for the alerts in **Discover** section.

ATTACK 1 – Credential Stuffing

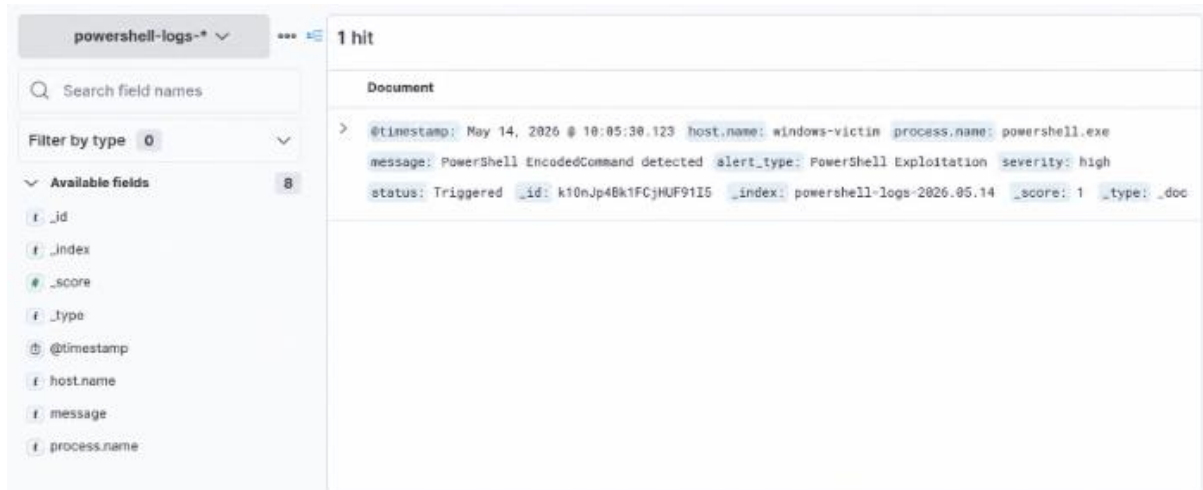


ATTACK 2 – DNS Tunnelling



```
alert_type: DNS Tunnelling attack message: Suspicious DNS query severity: high status: Triggered
_score: 1 _type: _doc
```

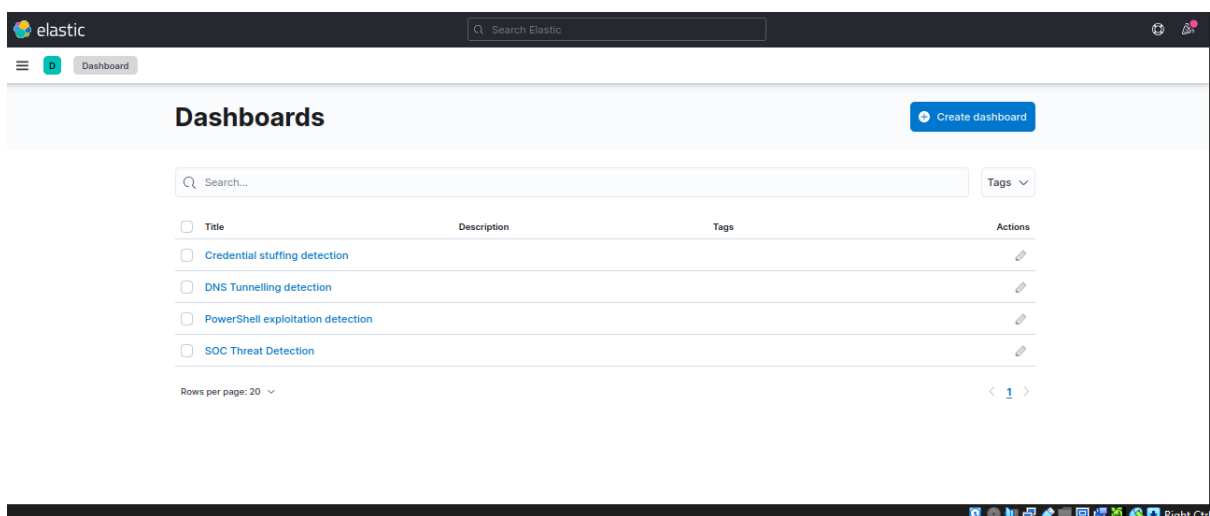
ATTACK 3 — PowerShell Exploitation

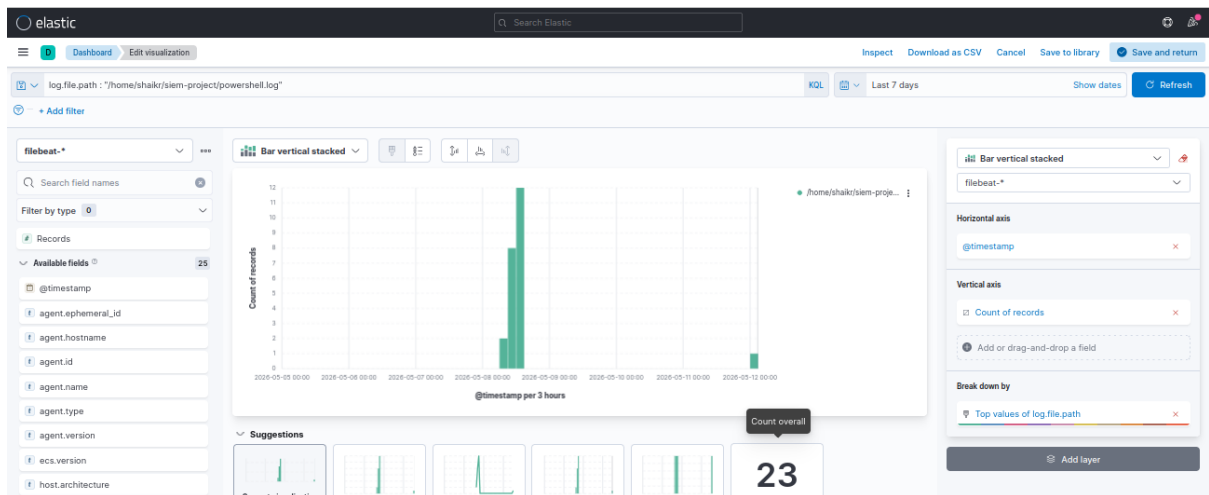


11. Creating dashboard

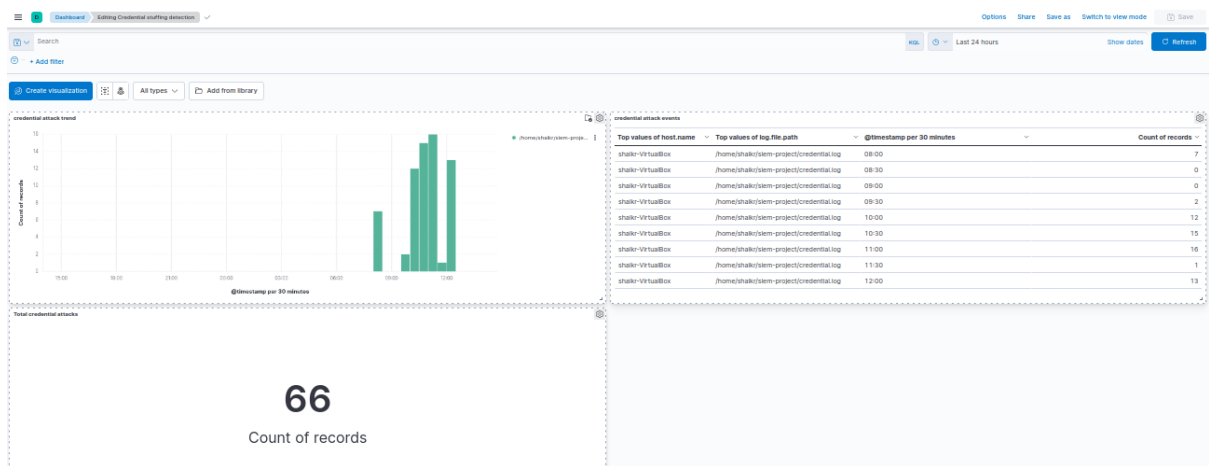
A visual representation of each attack about the total number of events over the time, a table representing the different fields involved and the graphical distribution of the trends.

Main Dashboard

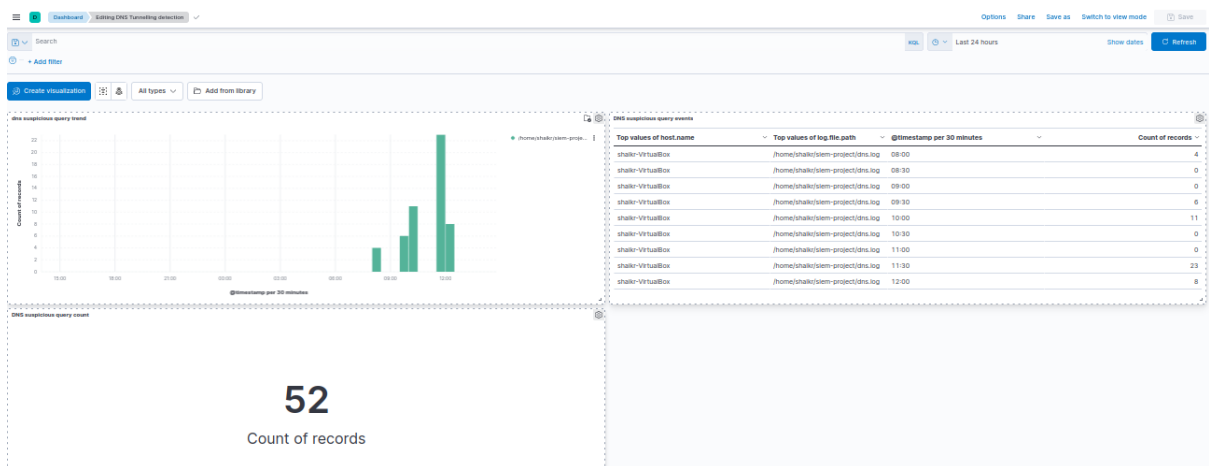




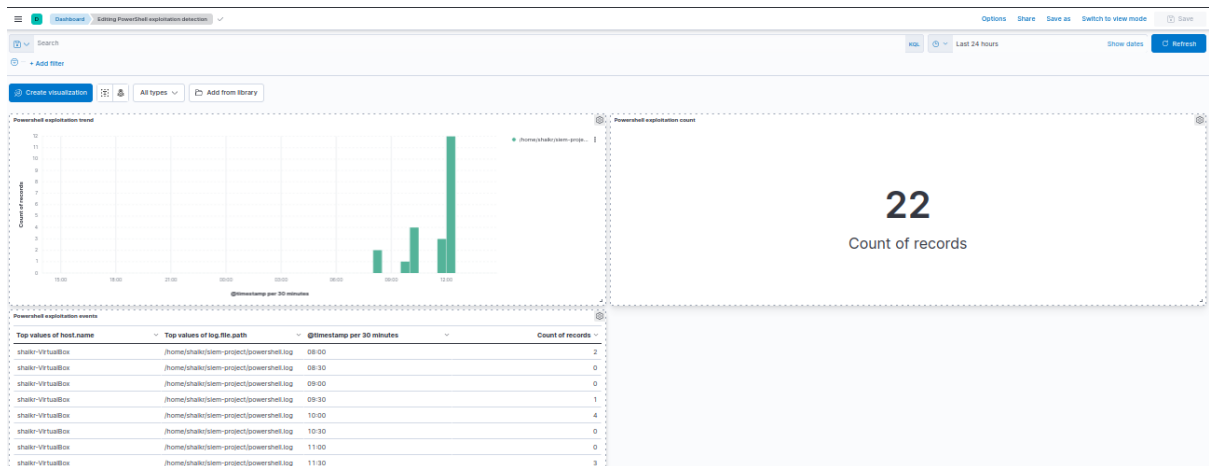
ATTACK 1 – Credential Stuffing



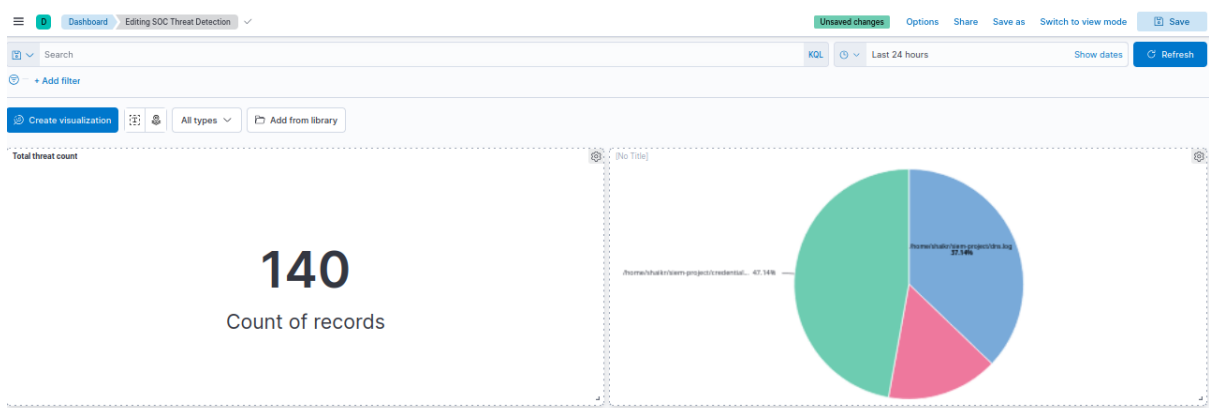
ATTACK 2 – DNS Tunnelling



ATTACK 3 – PowerShell Exploitation



Overall Attack Distribution Overview



12. Findings

- The ELK-based SIEM lab successfully analyzed simulated attack logs.
- Filebeat forwarded the logs to Elasticsearch where Kibana was used to query, visualize and create detection rules.
- Detection rules matched credential stuffing, DNS tunnelling, PowerShell attacks and helped in identifying attack patterns and triggered alerts.
- Dashboards provided a visual representation of threats, making understanding of attack distribution easier.

13. Mitigation Strategies

- Use MFA
- Use strong passwords
- Apply account lockout.
- Monitor for anomalies and unusual domains.
- Apply DNS filtering, EDR monitoring, PowerShell logging.
- Disable unrestricted execution

14. Conclusion

This project demonstrated how a SIEM solution can be used to detect suspicious activities by creating custom correlation or detection rules for credential stuffing, DNS tunnelling, and PowerShell exploitation. Using Elastic Stack, the simulated attack logs were successfully collected, analyzed, and visualized to understand attacker behavior. It provided practical exposure to security monitoring, alerting and dashboard creation, showing the importance of proactive threat detection.